

EEE 591 Machine Learning Project 1

As an engineer for the Acme Medical Analysis and Prediction Enterprises, I am tasked with developing an app to be used by doctors to aid in their prediction of heart disease. I have been studying and analyzing a database of patient data and outcomes from doctors and I need to build a predictor to determine whether a patient will get heart disease or not and give the confidence of that prediction. Many observations can be made from my analysis of the database. The variable “a1p2” is the absence or presence of heart disease, so that is the dependent variable because it’s what has to be predicted. All of the other variables like age, sex, resting blood pressure, maximum heart rate achieved, etc. are independent variables because they are used to predict the absence or presence of heart disease. After making a correlation matrix, I found that there are different types of correlations between many variables. Age has positive correlations with resting blood pressure (rbp), serum cholesterol (sc), exercise induced angina (eia), ST depression induced by exercise relative to rest (opst), and number of major vessels colored by fluoroscopy (nmvcf), which means that as age increases, so do all of these other variables. Sex also has positive correlations with thal and the target variable, a1p2, which suggests that there is a gender-related predisposition towards thal and the occurrence of heart disease. Chest pain type has positive correlations with eia, nmvcf, thal, and a1p2. Resting blood pressure is positively correlated with opst. Fasting blood sugar (fbs) and resting electrocardiographic results (rer) do not have any strong correlations with other variables. Maximum heart rate achieved has negative correlations with all variables except for fbs and highly negative correlations with age, cpt, eia, slope of the peak exercise ST segment (dests), and a1p2. Eia has positive correlations with cpt, thal, and a1p2. Opst has positive correlations with eia and a1p2 and high positive correlation with dests. The target variable, a1p2, has positive correlations with cpt, eia, opst, nmvcf, and thal. All of these correlations describe positive or negative relationships between variables and what an increase or decrease of a specific variable means for the other that it is being compared to. I also made a covariance matrix that provided some additional insights. Correlation matrices standardize the values of correlation from -1 to 1, so it is a bit easier to understand the positive and negative correlations between variables. Covariance matrices have different scales of measurement across variables and give information on the degree to which variables change together. Age and cpt have a covariance of 0.84 in the covariance matrix and that suggests that there is a strong positive relationship between the two, but the correlation matrix only showed a value of 0.097, which suggests a much weaker linear relationship, but that is due to the standardization of the values. The covariance between cpt and mhr is -0.32, which is a negative relationship and the correlation matrix shows a similar result. Based on all of these results, some of the variables that are not independent of each other are cpt and nmvcf and opst and dests because they have high correlation coefficients and that shows that there is a lot of dependence or strong relationship between them. The variables that are highly correlated with the target variable or the best predictors of heart disease are thal, nmvcf, eia, opst, dests, cpt, and mhr.

Part 2:

I performed various machine learning methods to explore different approaches possible to predict heart disease based on the dataset. By trying multiple methods, I wanted to understand how different models perform on the data and compare their accuracy to see which is the best method of predicting heart disease on this database. The machine learning methods I used were perceptron, logistic regression, decision tree learning, random forest, and K-Nearest Neighbor. For the Support Vector Machine, I used the Support Vector Classifier with a polynomial kernel. The accuracies of each model are:

Machine Learning Method	Prediction Percentage
Perceptron	57.41%
Logistic Regression	90.74%
Support Vector Machine	74.07%
Decision Tree Learning	81.48%
Random Forest	92.59%
K-Nearest Neighbor	70.37%

Random Forest had the highest accuracy of 92.59% among the models, which means it might be the most suitable approach for predicting heart disease based on the features in the dataset. Random forest combines multiple decision trees and reduces overfitting, which improves performance a lot. Logistic regression had the second highest accuracy of 90.74%. Perceptron had the lowest prediction percentage of 57.41%, which means it is the worst machine learning model to use for this dataset. The best value of k was chosen for the K-Nearest Neighbors algorithm by looping over different values and checking which value leads to the highest accuracy, but the highest accuracy only came out to 70.37%, which is not the best accuracy amongst the other models. I also tried other parameters for the algorithms like max depth for the decision tree classifier, for example, and these were the highest accuracy percentages I could come up with. In conclusion, the best machine learning model to predict the presence or absence of heart disease was Random Forest.